

Assignment 1

Assignment: Introduction to Web Development

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Answer:

Frontend, backend, and full-stack development are the three major parts of web application development.

Frontend Development

Frontend development is the part of a website or application that users can see and interact with. It focuses on the design, layout, buttons, forms, images, and overall user experience. Frontend developers mainly use **HTML, CSS, and JavaScript**.

Example:

When you open **Flipkart**, the homepage, product images, navigation bar, search box, and "Add to Cart" button are all part of the frontend.

Real-life

comparison:

Frontend is like the **showroom** of a shopping mall that customers visit and interact with.

Backend Development

Backend development works behind the scenes. It manages the server, database, authentication, and business logic. Backend developers commonly use technologies such as **Java, Python, PHP, Node.js, or C#**.

Example:

When you click "**Place Order**" on Flipkart, the backend checks product availability, processes payment, updates inventory, and stores your order details.

Real-life

comparison:

Backend is like the **warehouse and billing department**, where all the important processing happens.

Full-Stack Development

A full-stack developer works on both the frontend and backend. They can build an entire web application from the user interface to the database.

Example:

In a food delivery application like **Swiggy**, a full-stack developer can create the restaurant menu interface and also develop the system that manages orders, payments, and delivery tracking.

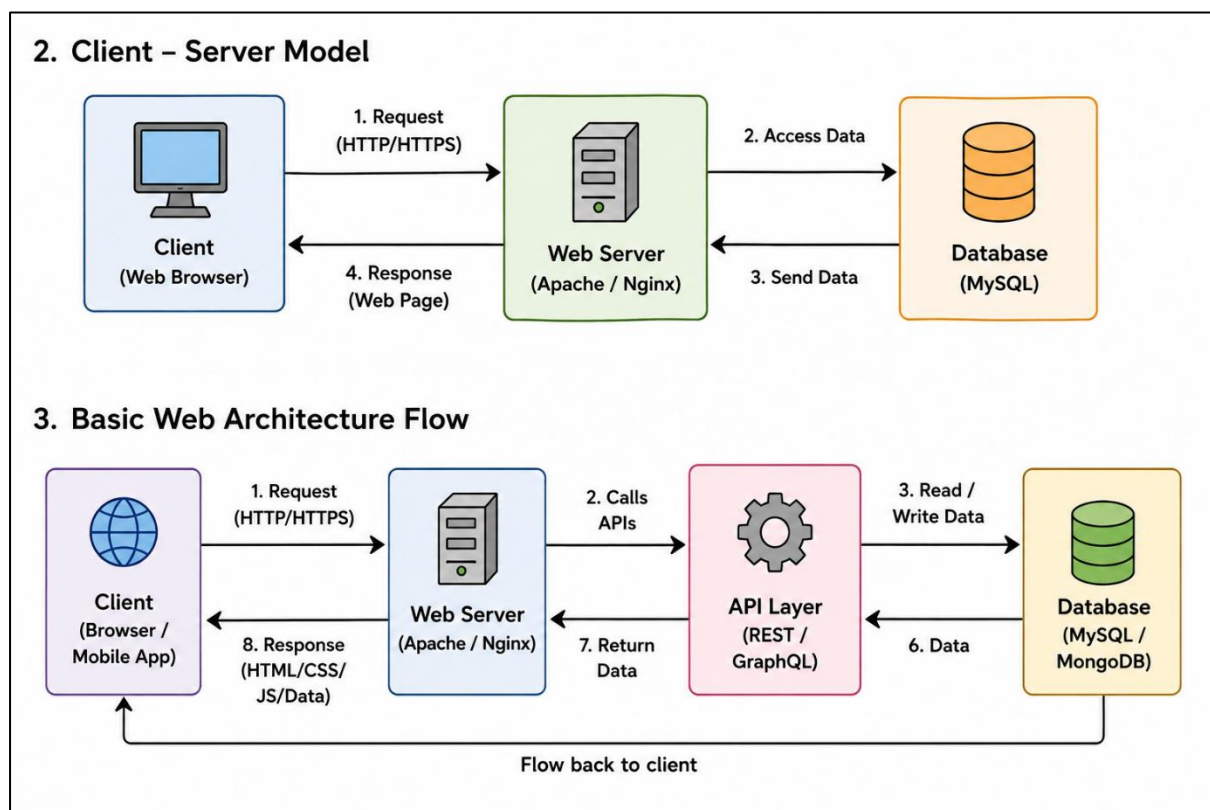
Real-life

comparison:

A full-stack developer is like someone who manages both the **customer-facing showroom** and **the warehouse**.

2. Create a simple diagram showing how the client-server model works in web architecture.

Answer:



3. Describe how a browser requests and displays a web page from a web server.

Answer:

Whenever a user enters a website address (URL) in a browser, several steps take place before the webpage appears.

1. The user types a website address such as www.example.com into the browser.
2. The browser sends an **HTTP/HTTPS request** to the web server.
3. The web server receives the request and processes it.

4. If required, the server communicates with the database to retrieve the requested information.
5. The server sends the webpage files (HTML, CSS, JavaScript, images, etc.) back to the browser.
6. The browser reads the HTML document first.
7. It then loads the CSS files for styling.
8. JavaScript is executed to make the webpage interactive.
9. Finally, the browser renders everything on the screen, allowing the user to interact with the website.

Example:

When you open **YouTube**, the browser requests the homepage, the server sends the webpage files, and the browser displays videos, buttons, menus, and recommendations.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Answer:

Tool	Purpose
Visual Studio Code (VS Code)	Used to write and edit HTML, CSS, JavaScript, and other programming languages.
Web Browser (Chrome, Edge, Firefox)	Displays and tests the developed website.
Live Server Extension	Automatically refreshes the webpage whenever code is saved.
Git	Tracks changes in source code and helps manage different versions of a project.
GitHub	Stores project code online and supports collaboration with other developers.
Node.js	Allows JavaScript to run outside the browser and is used for backend development.
npm (Node Package Manager)	Installs libraries and packages required for web development.

Tool	Purpose
Terminal/Command Prompt	Executes commands related to Git, Node.js, and project management.

5. Explain what a web server is and give examples of commonly used servers.

Answer:

A **web server** is software or hardware that stores website files and delivers them to users whenever they request a webpage through a browser.

The web server receives HTTP/HTTPS requests, processes them, and returns the appropriate webpage or data.

Commonly Used Web Servers

Web Server	Description
Apache Server	HTTP One of the oldest and most widely used open-source web servers.
Nginx	High-performance web server commonly used for handling heavy traffic.
Microsoft IIS	A web server developed by Microsoft for Windows-based applications.
LiteSpeed	Known for high speed and excellent performance, especially with WordPress websites.
Node.js Server	Frequently used to build modern JavaScript-based web applications.

Example:

When someone visits an online shopping website, the web server sends product pages, images, and other content to the user's browser.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Answer:

Frontend Developer

A frontend developer designs and develops the part of the application that users interact with. They ensure the website looks attractive, responsive, and easy to use.

Responsibilities:

- Design webpage layouts

- Build responsive interfaces
 - Add animations and user interactions
 - Improve user experience
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Backend Developer

A backend developer creates the server-side logic of the application.

Responsibilities:

- Develop APIs
 - Handle authentication
 - Process user requests
 - Manage business logic
 - Connect applications with databases
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Database Administrator (DBA)

A Database Administrator manages the database that stores application data securely and efficiently.

Responsibilities:

- Create and maintain databases
 - Perform backups
 - Improve database performance
 - Ensure data security
 - Manage user permissions
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7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Answer:

Installation Steps

1. Download **Visual Studio Code** from the official website.
2. Install VS Code by following the installation wizard.
3. Open VS Code.

4. Install the following extensions:
 - Live Server
 - Prettier – Code Formatter
 - HTML CSS Support
 - JavaScript (ES6) Code Snippets
5. Create a new project folder.
6. Add the following files:
 - index.html
 - style.css
 - script.js
7. Open index.html.
8. Right-click and select **Open with Live Server**.
9. The webpage will open in your default browser.

Note:

Take a screenshot showing:

- VS Code window
- Installed extensions
- index.html, style.css, and script.js
- Live Server running

(Attach your own screenshot here in the assignment.)

8. Explain the difference between static and dynamic websites. Provide an example of each.

Answer:

Static Website	Dynamic Website
Content remains the same for every visitor.	Content changes based on user interaction or database information.
Built using HTML and CSS.	Uses frontend, backend, and databases.
Faster because pages are pre-built.	Slightly slower because data is generated dynamically.
Easy to create and host.	More complex to develop.

Static Website

Suitable for portfolios and company websites.

Dynamic Website

Suitable for e-commerce, banking, and social media websites.

Example of a Static Website

A personal portfolio website that displays fixed information about the developer.

Example of a Dynamic Website

Amazon, Instagram, Facebook, or Flipkart, where content changes according to user accounts, searches, and preferences.

9. Research and list five web browsers. Explain how rendering engines differ between them.

Answer:

A web browser is software that allows users to access and view websites on the internet.

Common Web Browsers

Browser	Rendering Engine	Description
Google Chrome	Blink	Fast browser developed by Google.
Microsoft Edge	Blink	Microsoft's browser based on Chromium.
Mozilla Firefox	Gecko	Open-source browser known for privacy and customization.
Apple Safari	WebKit	Default browser for macOS and iOS devices.
Opera	Blink	Chromium-based browser with built-in VPN and extra features.

Rendering Engines

A **rendering engine** is responsible for reading HTML, CSS, and JavaScript code and displaying the webpage correctly.

- **Blink** (Chrome, Edge, Opera): Fast rendering, excellent support for modern web technologies, and widely used.
- **Gecko** (Firefox): Focuses on web standards, privacy, and independent browser development.

- **WebKit (Safari):** Optimized for Apple devices, offering good performance and energy efficiency.

Although browsers may display the same webpage, slight differences in rendering engines can cause variations in layout, animations, fonts, or CSS support. Therefore, developers test websites across multiple browsers to ensure a consistent user experience.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

